1. **New York Stock Exchange.** Suppose that the percentage returns for a given year for all Stocks listed on the New York Stock Exchange follows a *non*-normal distribution with mean $\mu = 11.2$ percent and a standard deviation of $\sigma = 8.4$ percent.

   (a) Given the tools we have learned in class, can you find the probability that a *single* stock has an annual return less than 34 percent? Explain why or why not. (Note: you do not need to calculate anything.)

   No. The problem states that the population follows a non-normal distribution. Thus, we cannot calculate this probability.

   (b) Consider drawing a random sample of $n = 5$ stocks from the population of all stocks and calculating the mean return of the five sampled stocks. Remember that we can think of the *observed* sample mean, $\bar{x}$, as a single draw from a distribution that relates to the random variable $\bar{X}$. We call this distribution *the sampling distribution of the sample mean.*

   i. What is the shape of the sampling distribution of the sample mean when n=5?
      A. Normal
      B. Non-normal
      C. Not enough information to tell

   ii. What is the mean of the sampling distribution of the sample mean when $n = 5$? (Report to two decimal places.)

   $$\mu_{\bar{X}} = \mu = \mathbf{11.2}$$

   iii. What is the standard error of the sampling distribution of the sample mean when $n = 5$? (Report to two decimal places.)

   $$\sigma_{\bar{X}} = \sigma/\sqrt{n} = 8.4/\sqrt{5} = \mathbf{3.76}$$

   iv. (**Yes** or **No**) Can you find now the probability that the mean return of the five sampled stocks is less than 34 percent?

   (c) In general a sample size of $n = 30$ should be sufficiently large to ensure that $\bar{X}$ follows an approximately Normal distribution. Thus, we are able to calculate probabilities associated with sample means even though we are drawing from a *non*-normal distribution using the Standard normal table. To answer the following questions you will want to use the following z-score formula:

   $$z = \frac{\bar{x} - \mu_{\bar{x}}}{\sigma} = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}.$$

   i. Specify the sampling distribution of the sample mean when $n = 30$, i.e. what are the mean, the standard error and the shape of the sampling distribution?

      A. shape: (Multiple Choice)
      - Normal

- Not Normal
- Not enough information to tell

B. mean: $\mu_{\bar{X}} = \mu = \mathbf{11.2}$

C. standard error: $\sigma_{\bar{X}} = \sigma/\sqrt{n} = 8.4/\sqrt{30} = \mathbf{1.53}$

ii. Find the probability that the mean return for the 30 sampled stocks is less than 10 percent.

Note that $SE(\bar{X}) = 8.4/\sqrt{30} = 1.53$. Then,

$$P(\bar{X} < 10) = P\left(Z < \frac{10 - 11.2}{1.53}\right) = P(Z < -0.78) = \mathbf{0.2177}$$

iii. Find the probability that the mean return for the 30 sampled stocks is between 10.5 and 14 percent.

$$\begin{aligned} P(11 < \bar{X} < 14) &= P\left(\frac{10.5 - 11.2}{1.53} < Z < \frac{14 - 11.2}{1.53}\right) \\ &= P(-0.46 < Z < 1.83) \\ &= P(Z < 1.83) - P(Z < -0.46) \\ &= 0.9664 - 0.3228 = \mathbf{0.6436} \end{aligned}$$

iv. Companies are often concerned with the notion of worst probable case performance of their portfolios. One figure a company might be interested in is what mean return will 99% of these portfolios of 30 stocks perform better than. Compute this value and report the answer to two decimal places.
This is asking about the 1st percentile, so the appropriate Z-score is -2.33. Then we have the following for X:

$$\begin{aligned} x &= z * \frac{\sigma}{\sqrt{n}} + \mu \\ &= -2.33 * \frac{8.4}{\sqrt{30}} + 11.2 = -2.33 * 1.53 + 11.2 \\ &= 7.64 \end{aligned}$$

2. **Critical Values from the t-table.**

(a) **Accurate to the nearest 3 decimal places**, what is the critical value ($t_{\frac{\alpha}{2},n-1}$ or $z_{\frac{\alpha}{2}}$) that corresponds to the given confidence levels and degrees of freedom. Fill in the following table with the appropriate critical values:

Confidence level	σ unknown ($t_{\frac{\alpha}{2},n-1}$)			σ known ($z_{\frac{\alpha}{2}}$)
	$df = 6$	$df = 30$	$df = 100$	
80%	1.440	1.310	1.290	1.280 / 1.282
90%	1.943	1.697	1.660	1.645
95%	2.447	2.042	1.984	1.960
99.9%	5.959	3.646	3.390	3.300 / 3.291

(b) As the degrees of freedom increase, how does the $t_{\frac{\alpha}{2},n-1}$ value compare to the $z_{\frac{\alpha}{2}}$ value?
As the degrees of freedom increase, the $t_{\frac{\alpha}{2},n-1}$ value gets closer to the $z_{\frac{\alpha}{2}}$ value. This is because the t-distribution approaches the Standard Normal distribution as the degrees of freedom increase.

(c) A 90% confidence interval using df = 6 is based on how many observations? 7

(d) A 95% confidence interval using df = 30 is based on how many observations? 31

(e) [**Don't forget how to use the Standard normal table**] What is the positive value of $z_{\frac{\alpha}{2}}$ for a 74% confidence interval when σ is known? 1.13

3. **Probabilities from the t-table.** Notice that some of the questions below use df and some use n. What is the value of

(a) $P(T < 3.326)$ for df = 14? 0.9975

(b) $P(T < -0.859)$ for n = 22? 0.2

(c) $P(T > 3.421)$ for df = 27? 0.001

(d) $P(T > 3.421)$ for n = 28? 0.001

(e) $P(T < -2.660)$ for n = 80? 0.005

(f) $P(|T| > 1.363)$ for df = 11? 0.2

4. Answer the following questions on-line by checking either **True or False.**

   (a) The t-distribution is characterized by the parameters μ (the mean) and σ (the standard deviation). False

   (b) The t-table (Table D) works with the area above (right or upper tail) while the z-table (Table A) works with the area below (left or lower tail). True

   (c) The t-distribution approaches the Standard Normal Distribution, $N(0, 1)$, as the degrees of freedom increase because s estimates σ with less precision when the sample size increases. False

5. A study was conducted among individuals with chronic pain to determine how effective acupuncture is in relieving pain. A random sample of 15 subjects were chosen and their sensory rates were measured. Assume that the measurements on sensory rates follow a normal distribution. The data from the study are given below:

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8.6  9.4  7.9  6.8  8.3  7.3  9.2  9.6  8.7  11.4  10.3  5.4  8.1  5.5  6.9
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   (a) What is the value of $\bar{x}$, the sample mean (Use SAS, R, JMP or MS Excel to calculate)? (Round your answer to the nearest 2 decimal places) $\bar{x} = 8.23$

   (b) What is the value of s, the sample standard deviation (Use SAS, R, JMP or MS Excel to calculate)? (Round your answer to the nearest 2 decimal places) $s = 1.67$

   (c) Report the degrees of freedom associated with the t-distribution. $df = 14$

   (d) What is the value of $t_{\frac{\alpha}{2},n-1}$ for the 95% confidence interval for the mean sensory rate? (The value obtained from Table-D, accurate to the nearest 3 decimal places) $t_{\frac{\alpha}{2},n-1} = 2.145$

   (e) What is the margin of error for the 95% confidence interval for the mean sensory rate? (Round your answer to the nearest 2 decimal places)
   $m = 2.145 \times \frac{1.67}{\sqrt{15}} = 0.92$

   (f) Calculate the 95% confidence interval for the mean sensory rate.

      i. What is the value of the lower bound of the 95% confidence interval? (Round your answer to the nearest 2 decimal places)
      $8.23 - 2.145 \times \frac{1.67}{\sqrt{15}} = 7.31$

      ii. What is the value of the upper bound of the 95% confidence interval? (Round your answer to the nearest 2 decimal places)
      $8.23 + 2.145 \times \frac{1.67}{\sqrt{15}} = 9.15$

   (g) Interpret the confidence interval you constructed above in the context of the problem.
   We are 95% confident that the true mean sensory rate for all individuals with chronic pain is between 7.31 and 9.15.

(h) Assume that the standard deviation of all the sensory rates is known to be 1.67. Calculate the 95% confidence interval for the mean sensory rate under this situation.

i. What is the value of the lower bound of the 95% confidence interval? (Round your answer to the nearest 2 decimal places)
$8.23 - 1.96 \times \frac{1.67}{\sqrt{15}} = 7.38$

ii. What is the value of the upper bound of the 95% confidence interval? (Round your answer to the nearest 2 decimal places)
$8.23 + 1.96 \times \frac{1.67}{\sqrt{15}} = 9.08$

(i) Is the 95% confidence interval constructed in part(f) **wider** or **narrower** than the 95% confidence interval in part(h)? Explain why.
The 95% confidence interval in part(f) is **wider** than the 95% confidence interval constructed in part(h). This is because we used the $t_{\frac{\alpha}{2},n-1}$ value of 2.145 which is higher than the $z_{\frac{\alpha}{2}}$ value of 1.96, to account for the additional source of uncertainty due to estimating σ with s.

6. [**8pts**.] **Stating the hypotheses.** In each of the following situations a significance test for a population mean μ is called for. Identify the null hypothesis H_0 and the alternative hypothesis H_A in each case. Make sure to use proper notation! (*e.g.* $H_0 : \mu = \mu_0$).

(a) [**2pts**.] 50 college students are selected to participate in a Kinesiology study. A researcher believes his workout regime increases body mass. All 50 participate in a weight lifting regime three times a week and their before and after body masses are measured.

$$H_0 : \mu \leq 0 \qquad \text{versus} \qquad H_a : \mu > 0$$

(b) [**2pts**.] Ten years ago there was a Mercury spill in Lake Tomahawk that caused serious damage to humans and wildlife. Today, researchers want to know whether it is safe for humans to use the lake recreationally. The EPA says water is safe if no more than 2 mg/L mercury is in the water.

$$H_0 : \mu \geq 2 \qquad \text{versus} \qquad H_a : \mu < 2$$

(c) [**2pts**.] A public accounting firm is auditing a very large corporation. One of the components the auditors need to check is the balance on the company's accounts receivable at fiscal year end. It would be very costly to reconcile every customer of the company and how much they owe. A partner at the accounting firm claims the average accounts receivable balance is $159,000. A junior associate at the accounting firm believes the partner is wrong, but would like to prove with statistical evidence.

$$H_0 : \mu = 159,000 \qquad \text{versus} \qquad H_a : \mu \neq 159,000$$

(d) [**2pts**.] The population mean IQ is 105. A psychologist wants to test the hypothesis that a new cognitive treatment increases the IQ of patients who undergo the treatment.

$$H_0 : \mu \leq 105 \qquad \text{versus} \qquad H_a : \mu > 105$$

7. [**10pts**.] **Walleye Trouble.** Walleye is a fish that is native to the northern United States and Canada that is known for its excellent taste. It is especially popular in Minnesota where it is deemed their state fish. As a result it is often overfished in Minnesota lakes, especially Lake Mille Lacs. One way to measure the health of the population is through the average length of the fish because fish grow their entire lives. The DNR considers the walleye population in a lake to be strong if the mean length of walleye in the lake is more than 17 inches. The DNR conducted a fish survey and netted 660 walleye and calculated an average length of 17.5 inches and a standard deviation of 4.68 inches. Is this sufficient evidence to say the walleye population in Lake Mille Lacs is strong?

(a) Conduct a hypothesis test of the above situation, using a significance level of α=0.05.

i. [**2pts**.]

$$H_0 : \mu \leq 17 \qquad \text{versus} \qquad H_A : \mu > 17$$

ii. [**1pt**.] $t = \frac{17.5-17}{4.68/\sqrt{660}} = 2.744708$

iii. [**1pt**.] Correct p-value is 0.009 using the given table

iv. [**2pts**.] This p-value is the probability, assuming the true mean length of walleyes in Lake Mille Lacs is 17 inches, of observing a test statistic t greater than 2.744708
1/2pt. for "p-value is the probability " , 1/2pt. for "of observing a test statistic t", 1/2pt. for " greater than or equal to 2.744708", 1/2pt. for saying "assuming the true mean length=17" or "H_0 is true"

v. [**1pt**.] p-value $= 0.009 < \alpha = 0.05 \Rightarrow$ reject H_0

vi. [**3pts**.] There is statistically significant evidence at the $\alpha = 0.05$ level to indicate that the walleye population in Lake Mille Lacs is strong.
1pt. for "There is statistically significant evidence", 2pt. for "to indicate that the walleye population in Lake Mille Lacs is strong"